

Detecting and Measuring *gclid*-Based Tracking on the Web

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Abstract

Online advertising relies heavily on URL-embedded identifiers such as the Google Click ID (*gclid*) for conversion attribution. As browsers and regulations increasingly restrict third-party tracking, advertisers have adapted by promoting *gclid* into the first-party context, transforming a transient URL parameter into a persistent first-party cookie that evades traditional tracking protections. Yet, *gclid*-based tracking across the open web and under different consent conditions remains poorly characterized.

In this paper, we collect Google Ads URLs (*AdURLs*) containing *gclid* from six vantage points (VPs) over two months and conduct ten measurement rounds. By interacting with consent banners, we measure *gclid* tracking from URL parameter to first-party cookie storage and third-party exfiltration. Among websites with consent banners, 51.2%–64.8% continue to set *gclid* cookies after explicit rejection, while only 4.5% correctly set them exclusively after acceptance. Even in Germany, only 30.9% of advertisers correctly honor consent. Across all VPs, 14.7% of advertisers persistently set the cookie regardless of consent state or location. Moreover, *gclid* non-compliance correlates with broader privacy violations, as advertisers disregarding *gclid* consent signals set post-rejection third-party tracking cookies at up to 77.6%, compared to 22.7%–51.0% among compliant sites. We further find that ten advertisers circumvent Safari’s Intelligent Tracking Prevention through parameter shadowing. Finally, despite rejection, advertisers leak *gclid* to a median of 3.0–4.6 third-party domains. Our findings reveal widespread consent violations and browser circumvention in *gclid*-based tracking.

CCS Concepts

• Security and privacy → Privacy protections.

Keywords

Conversion Tracking, Google Click Identifier, *gclid*, Cookie Consent, Consent Mode, Tracking Compliance, Parameter Shadowing, Link Decoration, Privacy, GDPR, Online Advertising, Measurement

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1 INTRODUCTION

In the digital era, online advertising becomes the central infrastructure of economic activities, where users search and conduct daily commercial transactions [26]. By delivering advertisements to the appropriate user segments, advertisers are able to increase the effectiveness of their campaigns and maximize returns. Through precise audience targeting, this ecosystem generates substantial revenue and has become the primary financial foundation of the modern web [21, 23].

To optimize advertising effectiveness, online advertising platforms increasingly rely on identifier-based mechanisms to link ad clicks with subsequent conversions. These identifiers enable advertisers to record click events, track browsing behavior, and attribute actions such as purchases, registrations, or form submissions. One widely used identifier is the Google Click ID (*gclid*), which operates within the Google Ads ecosystem. When a user clicks on an advertisement on a publisher website (e.g., a news site), Google Ads automatically appends *gclid* to the advertiser URL (*AdURL*). The *AdURL* is the click URL embedded in the advertisement, which typically initiates a sequence of redirects before reaching the advertiser’s landing page (see Figure 1). This mechanism associates the click event with subsequent conversion actions (e.g., purchases, sign-ups, or form submissions) on the landing page, enabling attribution of user actions to prior advertisement clicks. As a result, *gclid* plays a central role in the effectiveness of modern online advertising by supporting fine-grained attribution and performance measurement [14].

However, this mechanism introduces significant privacy risks. As browsers and regulatory frameworks increasingly restrict third-party tracking, advertisers have adapted by leveraging mechanisms such as *gclid* within the first-party context [13, 16, 37]. When the *gclid* parameter is appended to the *AdURL*, it is only available for that single request. To enable attribution beyond the landing page, advertisers store *gclid* in a first-party cookie so it can persist across subsequent page loads and sessions. This persistence enables advertisers long-term linkage of user activity. As a result, *gclid* can be used to build a broader behavioral history, linking multiple visits and interactions over time. Furthermore, if *gclid* is further shared with third-party tracking domains, this linkage extends across tracking domains. This enables cross-context tracking, where user activity can be correlated across different services and domains. Prior works have shown that identifiers embedded in URLs and

cookies can be repurposed for cross-site tracking and long-term profiling [13, 15, 39]. Despite the widespread adoption of such techniques across the ad-funded web, the behavior of *gclid*-based tracking remains insufficiently characterized, particularly under controlled consent conditions and across different geographical regions, motivating our systematic measurement study.

In this study, we conduct an empirical measurement of *gclid*-based tracking on the web. Our approach follows a three-stage measurement design. First, we perform a longitudinal collection of Google Ads URLs (*AdURLs*) containing *gclid* parameters from six geographic vantage points (§ 3.1). Second, we visit the corresponding advertiser landing pages under three controlled consent conditions (i.e., *no interaction*, *accept all*, and *reject all*) while capturing network requests and browser storage (§ 3.2). Third, we analyze *gclid*-based tracking by examining its persistence in cookie storage and its exposure in HTTP requests and responses (§ 3.3). Beyond these three stages, we present the first measurement of *gclid* parameter shadowing in the wild, where advertisers preserve *gclid* through an alternative parameter to evade browser-level filtering mechanisms (§ 3.4). Through this methodology, our study provides a systematic characterization of *gclid*-based tracking across the web under varying consent conditions, browser environments, and geographic locations.

Overall, our main findings are summarized as follows:

- Our measurements reveal that *gclid*-based tracking is pervasive across the web and largely independent of user consent. Across 1,412 advertisers and six geographic vantage points (VPs), a majority of advertisers (51.2%–65.3%) continue to set *gclid* as a first-party cookie even when users explicitly reject or do not interact with consent banners. Even in the presence of consent interfaces, only a minority of advertisers correctly restrict *gclid* storage based on user choice (§ 4.1).
- Across all six VPs, only 17.5%–30.9% of advertisers set *gclid* exclusively when consent is accepted, refraining from doing so under both rejection and no-interaction conditions. When considering only advertisers that exhibit the same behavior across all six VPs, only 4.5% are fully compliant, while 14.7% persistently set *gclid* regardless of consent condition. Furthermore, non-compliance extends beyond *gclid*: advertisers that ignore consent signals for *gclid* also exhibit the highest rates of third-party cookie setting after rejection (44.5%–77.6%), while advertisers that correctly handle *gclid* show substantially lower violation rates across both mechanisms (22.7%–51.0%) (§ 4.2).
- We identify 19 advertisers that preserve *gclid* in alternative URL parameters to mitigate browser-level tracking restrictions. Even when browsers such as Safari and Brave remove the original *gclid* parameter, these implementations allow the identifier to be recovered and, in some cases, stored as a first-party cookie. This demonstrates that *gclid*-based tracking can persist despite browser protections through the use of alternative parameter mechanisms (§ 4.3).
- We find that advertisers transmit *gclid* to multiple third-party domains, with a median of 3–5 receivers even when users reject or do not interact with consent banners. The majority of tracking domains (71.6%–79.7%) appear under

both acceptance and rejection conditions, indicating that most tracking is not dependent on user consent. Moreover, these domains are highly consistent across countries (Jaccard similarity 0.84–0.9), suggesting a globally shared tracking ecosystem (§ 4.4).

Finally, we discuss factors that may contribute to these behaviors, including implementation challenges in Google Consent Mode, and outline limitations of our study and directions for future work (§ 5).

2 BACKGROUND AND RELATED WORK

In this section, we present background on *gclid*-based tracking and Google Consent Mode, and discuss prior work on web tracking and cookie consent mechanisms.

2.1 Conversion Tracking

Conversion tracking in online advertising relies on URL-embedded identifiers such as the Google Click ID (*gclid*), which links user interactions from ad clicks to downstream conversion events. *gclid* is a unique identifier automatically appended by Google Ads to advertisement URLs, an *AdURL*, to link user clicks with subsequent actions. Similar identifiers are used across advertising platforms, including *fbclid* (Meta), *ttclid* (TikTok), and *msclkid* (Microsoft). These identifiers enable attribution of user interactions, such as product purchases or form submissions, to specific advertisements.

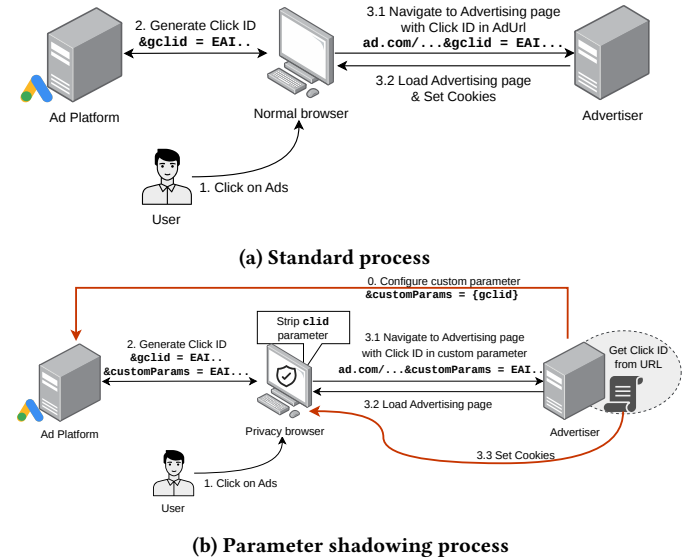


Figure 1: Conversion tracking process with *gclid*: (a) Standard tracking flow, where *gclid* is appended to the *AdURL* and stored as a first-party cookie; (b) Parameter shadowing, where *gclid* is duplicated into alternative parameters to bypass browser-level stripping and later restored by the advertiser.

Figure 1a shows the standard conversion tracking process. When a user clicks an advertisement (step 1), the advertising platform generates a unique *gclid* and appends it to the landing page URL (*AdURL*) (step 2). The browser then navigates to the advertiser's

website, where the *gclid* appears as a URL parameter (step 3.1). The advertiser stores this value in the browser, typically as a first-party cookie (step 3.2). When a conversion event occurs (e.g., a purchase), the stored *gclid* is sent back to the advertising platform, which matches it with the original click to enable attribution. In addition to this *online conversion* process, advertising platforms also support *offline conversion tracking*. In this setting, advertisers capture the *gclid* value and store it on their server (e.g., via form submissions or backend logging). When a conversion occurs later (e.g., offline purchase), the advertiser matches the event with the stored *gclid* and uploads the data to the advertising platform for attribution.

Modern browsers such as Safari, Firefox, and Brave introduce tracking protections that remove known tracking parameters, including *gclid*, from an *AdURL* [4, 10]. As a result, the identifier may be stripped before reaching the advertiser’s website, potentially preventing conversion attribution. To address this limitation, advertisers may adopt a technique known as *parameter shadowing*, shown in Figure 1b. In this approach, advertisers configure a custom parameter in the ad URL template that duplicates the *gclid* value (e.g., `&customParams={gclid}`) (step 0). When the ad is clicked (step 1), the *gclid* is appended to both the standard parameter and the custom parameter (step 2). While browsers may remove the known *gclid* parameter, the custom parameter remains, as it is not recognized as a tracking identifier. The browser then navigates to the advertiser’s page with the *gclid* value embedded in the custom parameter (step 3.1–3.2). Finally, the advertiser extracts the value from this parameter and stores it in a cookie or backend system (step 3.3), restoring the identifier for conversion tracking.

2.2 Google Consent Mode

Google Consent Mode [28] is a framework that allows website operators to communicate user consent choices to Google tags (e.g., Google Analytics, Google Ads, and the Conversion Linker), enabling these tags to adjust their behavior based on the user’s consent state. Consent Mode does not provide a consent banner itself; Instead, it integrates with an external consent management platform (CMP), which collects user choices and forwards them to Google’s measurement infrastructure. Importantly, Consent Mode is opt-in, requiring explicit configuration by advertisers for consent signals to take effect.

Consent Mode supports two implementation modes: *basic* and *advanced*. In the basic mode, Google tags are blocked until the user interacts with the consent banner, and no data is sent to Google beforehand. If consent is denied, no data is transmitted. In contrast, the advanced mode allows Google tags to load immediately, before user interaction. Under denied consent, tags send limited *cookieless pings*, which include metadata such as timestamps, user agents, and consent state, as well as whether the current page URL contains ad-click information such as *gclid*. Full measurement data and advertising cookies are only enabled once consent is granted. In Advanced Consent Mode, Google tags execute unconditionally upon page load, before user interaction. When consent is denied, the tags refrain from reading or writing persistent local storage (cookies) and instead transmit stateless network requests known as *cookieless pings*. Google leverages these basic, non-identifiable telemetry signals within machine learning frameworks to perform

conversion modeling and behavioral analysis. By algorithmically estimating unobserved conversions lost to consent refusals or browser restrictions, this mechanism bridges measurement gaps and preserves data utility while aiming to maintain user anonymity and regulatory compliance.

The framework defines four consent parameters: *ad_storage*, *ad_user_data*, *ad_personalization*, and *analytics_storage*. Among these, *ad_storage* controls whether advertising cookies, including those derived from *gclid*, may be stored on the user’s device. When set to denied, advertising cookies should not be set. However, even in this state, cookieless pings may still indicate the presence of *gclid* in the page URL. Advertisers must explicitly enable the *ads_data_redaction* setting to suppress this information [28].

To enforce consent-aware behavior, advertisers must invoke the consent API to declare default values before any tag execution. If this step is omitted, no consent state is applied, and Google tags execute without restriction, regardless of user interaction with the consent banner.

2.3 Related Work

Cookie consent banner measurement. The use of cookies by tracking ecosystems raises significant privacy concerns, as they enable persistent collection and aggregation of user data across websites [1, 36]. In particular, third-party cookies and cookie syncing mechanisms allow different domains to exchange identifiers, enabling the reconstruction of detailed cross-site behavioral profiles over time [2, 13, 39]. These practices often occur with limited transparency, making it difficult for users to understand how their data is collected, shared, and combined across the advertising ecosystem [32, 45, 47]. As a result, such large-scale tracking increases the risk of intrusive profiling and loss of anonymity.

In response to these concerns, regulations such as the General Data Protection Regulation (GDPR) [22] and the ePrivacy Directive [8] impose stricter requirements on transparency and user control. In particular, these regulations require websites to obtain informed user consent before storing or accessing non-essential cookies. In practice, this requirement is implemented through cookie consent banners, which serve as the primary interface for collecting and managing user choices. Cookie consent banners allow users to accept, reject, or customize their privacy preferences regarding data collection. However, prior work shows that these mechanisms are often ineffective in practice. They may obstruct access to content or employ interface designs that influence user decisions, thereby limiting meaningful user control [38, 49]. In addition, websites use tricks such as using long privacy terms [33], designing user interfaces to navigate users to consent action [12, 33, 38], or designing opt-out settings that are difficult to find or misleading [31]. This navigates users to choose the “accept all” in the cookie consent [9]. Furthermore, studies [20] and [44] consistently showed that ignoring the cookie banner provides an incomplete view of the web and that web tracking becomes significantly more prevalent after consent is given. Furthermore, accepting consent results in a higher number of third-party cookies being set compared to other options.

Web tracking measurement. Web tracking refers to the collection and analysis of users’ online behavior, enabling websites and third parties to monitor browsing activities across sessions and platforms.

Following studies [1, 16, 18–20] have shown that web tracking is widely used and have demonstrated that it can undermine browser cookie policies. Another method of tracking users is *cookie syncing*, which allows advertising platforms to synchronize user identifiers stored in cookies, enabling them to track users across different websites and enhance their cross-site tracking capabilities. Previous studies [1, 2, 13, 39] have shown that third-party trackers exploit first-party cookies to track users, even reconstructing a significant portion of users' browsing histories.

Furthermore, many studies have confirmed that web tracking exposes sensitive user data to trackers. Falahrestegar et al. [24] highlighted how organizations engage in extensive ID-sharing to track users across the web, regardless of whether they are logged in or out. In 2021, Dao et al. [15] revealed that over 42% of shopping sites leak personally identifiable information during sign-in, enabling a persistent tracking technique that identifies users across different platforms and devices. Pham et al. [41] demonstrated that web trackers utilize both extensive ID-sharing and PII leaks during Single Sign On authentication flows to persistently track users.

Recently, Dao et al. [17] measured ad clicks and *glid* in YouTube's advertising ecosystem across multiple browsers and geographic locations. While their work characterized the tracking mechanism itself, our work addresses two complementary dimensions that their study does not: advertiser compliance with consent signals, and the evasion of browser-level tracking protections. Concretely, our study differs and extends in the following ways: (1) shifting the focus from tracking mechanism characterization to advertiser compliance, by measuring *glid* behavior under three distinct consent conditions and systematically classifying non-adherence to consent signal across six VPs; (2) measuring *glid*-based tracking as it occurs in the context of display advertising on news websites, rather than video advertising on a single platform; (3) linking first-party *glid* handling to third-party tracking cookie behavior under consent rejection, revealing that compliance in one tracking mechanism does not imply broader privacy-protective behavior; (4) and providing the first characterization of *glid* parameter shadowing, where advertisers alias the identifier under alternative parameters to evade browser-level tracking protections such as those deployed by Safari and Brave.

3 METHODOLOGY

In this section, we present our end-to-end measurement pipeline to study *glid*-based tracking across the web. Our approach consists of three components as shown in Figure 2: (i) large-scale *AdURLs* collection across multiple geographic VPs, (ii) controlled visits to advertiser pages under different cookie banner interaction conditions, and (iii) analysis of *glid* propagation across cookies and HTTP request/response.

3.1 *AdURLs* Collection

To systematically study *glid*-based tracking in the wild, we first construct a dataset of *AdURLs* that contain embedded *glid*. Our goal is to capture *AdURLs* as they are presented to users under realistic browsing conditions, while maintaining control over consent interactions and measurement consistency across VPs. To this end, we design an automated crawling framework that simulates

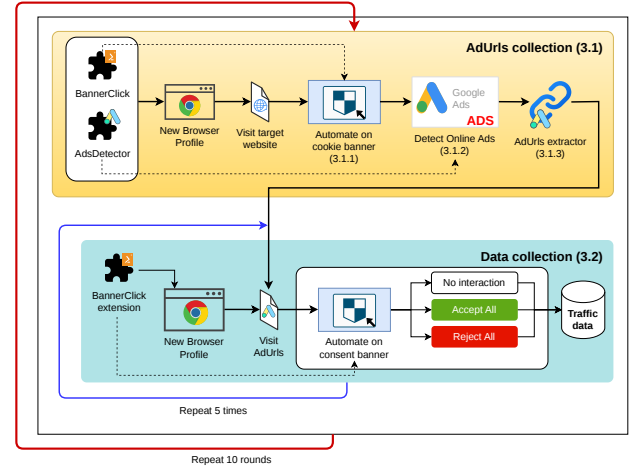


Figure 2: Overview of our data collection for a single crawling round. The top panel illustrates *AdURLs* collection, while the bottom panel shows data collection under three consent conditions, with five runs per condition. This process is repeated for ten rounds over two months, with measurements conducted across six geographic VPs in each round.

user visits to websites and collects advertisements displayed during page loads. The crawler operates in a controlled browser environment and performs three main steps: (1) detecting and interacting with consent banners; (2) identifying and capturing online advertisements rendered on the page; (3) extracting the corresponding *AdURLs*, with particular focus on those containing *glid* parameters.

3.1.1 Automate banner detection and interaction. Through manual inspection, we find that improper handling of cookie banners can obstruct page functionality and prevent advertisements from loading. In many cases, advertisements are displayed only after users interact with the banner (e.g., by accepting cookies). As a result, failing to interact with cookie banners can limit the visibility of advertisements.

To systematically detect and interact with these banners in a consistent and automated manner, we integrate and extend *BannerClick* [43], a tool that automatically detects and interacts with cookie consent banners. *BannerClick* achieves high accuracy (around 99%) in automating banner detection and interaction and supports a wide range of banner implementations. We further evaluate its performance through manual verification of both acceptance and rejection interactions, achieving accuracies of 94% and 90%, respectively (see Appendix A). While *BannerClick* was originally designed for Firefox, we extend its implementation to support Chromium-based browsers, enabling consistent cross-browser interaction across VPs.

On each page visit, our crawler invokes *BannerClick* immediately after page load to detect a cookie banner and perform the configured interaction. Depending on the experimental setup, the system simulates one of three behaviors: no interaction, explicit acceptance, or explicit rejection of cookies. This allows us to capture *glid*-based tracking behavior under different consent conditions in a controlled and reproducible manner.

3.1.2 Online Advertisement detection. We develop a browser extension, *AdsDetector*, to automatically detect online advertisements on websites. The extension leverages widely used filter lists, including EasyList[5] and EasyPrivacy [6], which define rules for identifying advertising and tracking-related content. By relying on these lists, our system adopts a well-established definition of what constitutes an advertisement.

The detection process begins when a webpage loads, at which point *AdsDetector* retrieves and parses the filtering rules. These rules include blocking rules, exception rules, and domain-based patterns. Following *Adblock Plus filters explained* [3], we convert these rules from their original textual format into CSS selectors, effectively translating filter rules into element-matching patterns that can be applied directly to the Document Object Model (DOM). *AdsDetector* then scans the webpage by matching these CSS selectors against DOM elements. When an element satisfies a rule, the extension marks it as an advertisement and annotates it directly in the DOM (e.g., by tagging or highlighting the element). These annotations are subsequently used to extract advertisement-related information, such as AdURLs, in later stages of our pipeline.

To account for dynamically loaded content, *AdsDetector* continuously monitors changes in the DOM to detect advertisements that appear after the initial page load. When new elements are inserted (e.g., through infinite scrolling or lazy loading), the extension incrementally scans only the newly added elements rather than reprocessing the entire page.

3.1.3 AdURLs extraction. After the extension detects and annotates advertisement-related elements within the DOM, we focus on systematically retrieving metadata and associated AdURLs from these elements without triggering user interaction events.

For each detected element, we first extract positional information, such as its bounding rectangle coordinates. We then scroll the browser viewport to bring the target element into view. This step is necessary because certain dynamic web elements only fully render or become accessible when visible within the viewport. We implement a multi-stage extraction strategy. In the first stage, we perform a direct scan of anchor (<a>) elements contained within the annotated node. Our method automatically collects all available href attributes and appends them to a result set, capturing explicit hyperlinks embedded within the advertisement container.

In the second stage, we address the common case where advertisements are encapsulated within iframe elements. For each iframe, we repeat the link extraction process by querying for anchor elements within its document. To ensure comprehensive coverage, we apply a recursive strategy to handle nested iframes, allowing us to extract AdURLs from deeply embedded advertisement structures.

Upon completion of the extraction process, all collected AdURLs are stored for further analysis. For verification purposes, we additionally capture screenshots of the detected advertisement elements, enabling manual inspection and validation of the extraction results. To assess the effectiveness of our approach, we manually verify a subset of extracted AdURLs and confirm that our method reliably captures advertisement landing URLs across a variety of webpage structures (see Appendix B).

3.1.4 AdURLs Collection Setup.

Target website selection. We first select websites for large-scale AdURLs collection. The selected websites should contain a high volume of online advertisements and maintain topical consistency to reduce noise during crawling. Following prior work [11], we focus on popular websites in the *News* category. We obtain the top 4,000 domains from the Tranco list generated on November 24, 2025¹, and categorize these domains. VirusTotal² is an online security service that analyzes IP addresses, domains, and URLs using multiple antivirus engines and security tools to identify potential threats and provide security-related information. In this study, we use VirusTotal to obtain security-related metadata for the collected domain URLs, including their categories. Based on this information, we identify 216 domains classified as *News*, which are selected for our measurement.

Location diversity. To capture geographic variation in advertising delivery, we measure from six VPs using Proton VPN³ to emulate user locations in Japan (Tokyo), the United States (California), Brazil (São Paulo), Australia (Adelaide), South Africa (Johannesburg), and Germany (Frankfurt).

AdURLs crawl process. For each measurement, we launch a fresh browser instance with no existing browsing history or stored cookies, ensuring a clean and controlled environment. We automatically install the *AdsDetector* extension in the browser.

We then navigate to a target website selected from our dataset of 216 domains in the *News* category. After the page loads, we use *BannerClick* to automatically detect and accept the cookie consent banner to ensure consistent page rendering and maximize the visibility of advertisements during crawling. We perform automated scrolling to trigger dynamically loaded content, simulating a lazy-loading behavior. Following prior work [20, 25, 35, 40, 41, 46], we set a timeout of 60 seconds per page to allow sufficient time for content and advertisements to fully load.

During this process, *AdsDetector* continuously scans the page and identifies displayed advertisements. For each detected advertisement, we use the AdURLs extractor to retrieve and store the corresponding AdURLs. In addition to the homepage, we visit three inner pages per website and repeat the same collection procedure, each with a 60-second timeout. To improve robustness, we repeat the collection process five times per website. To capture the dynamics of online advertising, we adopt a longitudinal crawling strategy over two months. We repeat the AdURLs crawl process across ten rounds. In each round, we execute 28 browser instances in headless mode to crawl all 216 websites across six VPs. This setup allows us to capture newly emerging advertisements over time and account for the dynamic nature of the ad ecosystem. We perform all our crawls from a server connected to the Internet through an institutional network, using an Intel Xeon Gold 5317 CPU and 256GB of memory and running Ubuntu 22.04.5 LTS.

The overview of our dataset is described in Table 1. Through ten rounds, across six VPs, we collect a total of 307,835 unique AdURLs. Among them, 80,301 unique AdURLs belong to Google and

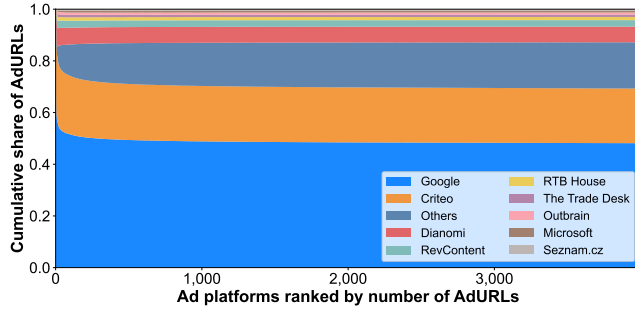
¹<https://tranco-list.eu/list/LYVN4>

²<https://virustotal.com>

³<https://protonvpn.com/>

Table 1: Overview of the AdURLs dataset.

Round	Date	# AdURLs	# AdURLs w/ <i>gclid</i>
1	May 14, 2026	30,264	6,219
2	May 30, 2026	33,986	8,825
3	June 05, 2026	34,157	8,130
4	June 11, 2026	30,853	8,037
5	June 15, 2026	36,667	14,023
6	June 19, 2026	36,318	9,016
7	June 24, 2026	34,041	8,150
8	June 29, 2026	35,412	6,906
9	July 05, 2026	29,755	4,739
10	July 12, 2026	32,849	6,256
Total (unique)		307,835	80,301

**Figure 3: Cumulative share of AdURLs by advertising platform, ranked by number of AdURLs.**

bring *gclid* in AdURLs. Note that not all Google AdURLs contain a *gclid* parameter; it is only included when the advertiser has enabled auto-tagging [27]. Besides AdURLs from Google, our crawler also collects AdURLs from other advertising delivery platforms, such as Criteo, Dianomi, and RevContent. We summarize the distribution of Advertising delivery platforms in Figure 3, shows that Google overwhelmingly dominates the ads ecosystem, with a substantially larger share than other platforms, highlighting a highly centralized advertising landscape.

3.2 Data Collection

After collecting AdURLs, we conduct experiments by visiting advertiser landing pages through these URLs. As AdURLs are dynamically generated and tied to specific advertising campaigns, they may change over time or become invalid (e.g., when campaigns are modified or discontinued), leading to broken or altered redirect chains [7, 30]. To minimize the impact of such temporal variations, we perform data collection immediately after each AdURLs crawling round, ensuring that measurements are conducted on fresh and valid AdURLs.

To reduce redundancy, we deduplicate AdURLs based on their destination advertiser. Specifically, we extract the final landing URL of each AdURL using the method described in § 3.3.1. If multiple AdURLs redirect to the same advertiser and differ only in parameter

values, we treat them as equivalent and retain a single representative AdURLs. Using this approach, we identify 1,502 unique advertisers, corresponding to 1,762 representative AdURLs, reduced from an initial set of 80,301 AdURLs containing *gclid*. The number of AdURLs exceeds the number of advertisers because a single advertiser may be associated with multiple structurally distinct AdURLs. These differences arise when advertisements are delivered through different Google domains (e.g., doubleclick.net, googleservices.com, and googlesyndication.com) or use different URL formats. We retain these distinct AdURLs to capture variations in *gclid*-based tracking behavior across different delivery paths.

To perform our analysis, we run a separate crawler using Selenium and Google Chrome 116 to automatically visit advertiser pages through AdURLs. For each visit, we launch a fresh browser instance without prior browsing history or stored cookies, ensuring a clean and controlled environment. We then open an AdURL and use *BannerClick* to detect the presence of a cookie consent banner and perform one of three actions: *accept all*, *reject all*, or *no interaction*. Next, reload the page and wait an additional 60 seconds. Overall, each page visit is bounded by a total timeout of 120 seconds, consisting of 60 seconds before interaction and 60 seconds after the page reload.

During this process, the crawler performs controlled scrolling to trigger dynamically loaded content. Throughout the entire process, both before and after the interaction, we collect HTTP request and response data, as well as cookie storage, to analyze *gclid*-tracking behavior. For each AdURL, we repeat the experiment five times under each consent condition to reduce variability and ensure consistent measurements. We conduct this data collection process across all AdURLs at six geographic VPs for each crawling round. The overview of the data collection process is shown in Figure 2

Note that we intentionally avoid direct interaction with advertisement elements (e.g., clicking on ads) and instead navigate to advertiser pages using the extracted AdURLs, enabling controlled and reproducible measurements across VPs and consent conditions.

3.3 *gclid*-based Tracking Detection

In this section, we explain a method to detect the *gclid* identifier and its exposures through a multi-stage pipeline. Overall, our method extracts *gclid* from AdURLs and traces their propagation across HTTP requests, responses and storage mechanisms.

3.3.1 Extract *gclid*. We parse each AdURL and extract all query parameters. From these parameters, we identify the *gclid* parameter and use its value as the target identifier for subsequent analysis. For example, an AdURL include a *gclid* parameter:

```
https://www.googleadservices.com/pagead/acclk?...
&gclid=EAIaIQobChMI...oTA0CB1HgxE3EAEYASAAEgJPUPD_BwE
&adurl=https://www.advertiser.com/
```

3.3.2 Detect *gclid* cookies. After extracting the *gclid* value for each AdURL, we trace its presence in cookie storage across browsing sessions. The cookie dataset includes cookie names, values, and associated domains.

For each record, we examine whether the cookie value contains the *gclid* identifier. We then classify cookies as first-party or third-party based on the relationship between the cookie domain and the

advertiser domain. A cookie is classified as first-party if its domain matches or is a subdomain of the advertiser’s domain; otherwise, it is classified as third-party.

3.3.3 Detect *gclid* exposures. We query our dataset to retrieve network requests associated with each browsing session. The network data includes URIs, headers and payload of HTTP requests/responses. We filter these datasets to find any occurrence of the *gclid* value.

We then scan multiple channels in each network interaction to detect exposures. Specifically, we query whether *gclid* appears in request URLs, payloads, and HTTP headers (e.g., referer, cookie). For each match, we record the domains that received *gclid*, and the channel where the exposure occurs. To identify third-party exposures, we compare each receiving domain with the domain of the advertiser. To this end, we extract the domain of the URI request, then classify this domain as first-party if it matches the visited web page (advertiser) domain. Otherwise, we classify it as third-party domains.

Finally, we determine whether a receiving domain corresponds to a tracking entity. We use the EasyList and EasyPrivacy blocklists to identify requests associated with tracking or advertising services. Specifically, we match request URLs against the rules in these blocklists to identify tracking-related resources. To assess whether a request would be blocked by these lists, we apply the blocklist rules directly to all collected requests across multiple runs and VPs.

3.4 *gclid* Parameter Shadowing Detection

Based on the parameter shadowing technique described in Figure 1b, we detect advertisers that conceal *gclid* within alternative URL parameters to bypass tracking protections. For each *AdURL*, we apply the same parameter extraction method as in § 3.3.1, searching for the *gclid* value across all remaining parameters to identify aliases.

To evaluate whether shadowing effectively circumvents tracking protections in practice, we visit the *AdURLs* using Safari 26.3 and Brave [10, 50], that implement protections against URL-based tracking parameters. These browsers strip known identifiers such as *gclid* from navigated URLs while leaving unrecognized alias parameters unaffected. For Safari, we perform experiments under both Default and Private Browsing modes. Private Browsing enables advanced tracking and fingerprinting protections by default [4], allowing us to isolate the effect of parameter stripping. For each advertiser who is using the *gclid* parameter shadowing technique, we visit the landing page via the corresponding *AdURL* under each browser configuration. We trace outgoing HTTP requests and responses to determine whether the shadowed parameter is recovered server-side and used to set a first-party cookie.

Note that, while the resulting behavior, namely the persistence of *gclid* as a first-party cookie, can be detected using the automated measurement pipeline described in § 3.2, we use manual experiments to validate that the shadowed parameter survives browser-level stripping and is subsequently extracted server-side to restore *gclid*-based tracking.

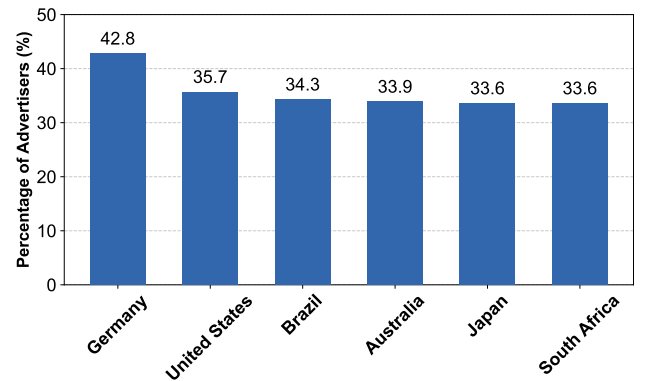


Figure 4: Percentage of advertisers (n=1,412) that display a cookie banner across VPs.

4 RESULTS

In this section, we present the results of our measurement study on *gclid*-based tracking across consent conditions and locations.

4.1 Overview

From our dataset, we analyze 1,412 advertisers across six VPs. While the deduplicated dataset contains 1,502 unique advertisers (see § 3.2), a small fraction is excluded from the final analysis. We exclude advertisers for which measurements cannot be consistently completed across VPs and consent conditions. This includes cases where 62 advertiser pages are blocked by bot detection mechanisms, as well as 28 cases where consent banner interactions cannot be reliably executed (e.g., only one option is available or interactions fail). Given that automated banner detection and interaction are not perfect (see Appendix A), we restrict our analysis to advertisers for which all consent conditions can be consistently applied, ensuring fair and comparable measurements. In total, 90 advertisers (approximately 6%) are excluded⁴. We break down the percentage of advertisers displaying the cookie consent banner in each country in Figure 4.

Our results show that only a minority of advertisers display a cookie consent banner in the observed VPs. Out of 1,412 advertisers, Germany has the highest adoption rate, at 42.8% (605 advertisers), followed by the United States at 35.7% (504 advertisers), a difference of 7.1 percentage points. This finding is consistent with previous studies, which suggest stronger compliance in Germany and Europe more broadly, potentially due to the strict data protection requirements imposed by the GDPR [44, 45]. The remaining countries show relatively similar adoption rates. Brazil has an adoption rate of 34.3% (484 advertisers), followed by Australia at 33.9% (478 advertisers), while Japan and South Africa have the lowest rates, both at 33.6% (475 advertisers). The adoption rates among these four countries differ by less than one percentage point. Overall, Germany exhibits the highest adoption rate, whereas the other

⁴We confirm that *AdURLs* from different Google domains (e.g., doubleclick.net, googleadservices.com, and googlesyndication.com) do not exhibit differences in *gclid*-based tracking behavior across these delivery paths.

countries show relatively limited variation in cookie consent banner adoption.

We then examine whether advertisers with and without these banners differ in how they store *gclid* as a first-party cookie, and present the breakdown in Table 2. Most advertisers set the *gclid* as a first-party cookie at very high rates. More details, about 72.2% to 80.3% of advertisers set *gclid* as a first-party cookie when no banner is shown. These percentages are higher when advertisers have cookie consent banners and users accept to allow advertisers to store cookies. Once users accept consent, from 80.6% to 84.1% of advertisers set *gclid* as a first-party cookie. This indicates that advertisers set *gclid* as an important tracking cookie, regardless of whether or not a banner consent cookie is present. Furthermore, even when users reject or do not interact, a majority of advertisers still set *gclid* as a first-party cookie. More detail, from 51.2% to 64.8% of advertisers set a cookie when the user rejects consent, and from 51.2% to 65.3% of advertisers set a cookie when the user does not interact with the consent banner. This raises concerns about the meaningfulness of user consent.

Table 2: Breakdown of advertisers that set *gclid* as a first-party cookie, grouped by cookie banner interaction.

Country	Banner Presence			No Banner
	Accept All	Reject All	No Interaction	
Australia	402 / 84.1%	306 / 64.0%	310 / 64.9%	738 / 79.0%
Germany	500 / 82.6%	310 / 51.2%	310 / 51.2%	583 / 72.2%
Brazil	390 / 80.6%	293 / 60.5%	296 / 61.2%	703 / 75.8%
Japan	394 / 82.9%	308 / 64.8%	310 / 65.3%	752 / 80.3%
United States	415 / 82.3%	323 / 64.1%	325 / 64.5%	721 / 79.4%
South Africa	389 / 81.9%	295 / 62.1%	295 / 62.1%	734 / 78.3%

Key takeaway: *gclid* is widely set as a first-party cookie across advertiser websites and locations, and persists even when users reject or do not interact with cookie consent banners, suggesting limited effectiveness of consent mechanisms.

4.2 Quantification of Non-Adherence to Consent Signals

To assess how advertiser behavior aligns with action on the consent banner, we classify advertisers into three categories based on how they handle the *gclid* identifier under different consent conditions across VPs (see Figure 5). Here, **yes** indicates that the advertiser sets the *gclid* identifier as a first-party cookie under the corresponding consent condition, whereas **no** indicates that no *gclid* cookie-setting behavior is observed. We aggregate observations across all crawl rounds and repeated runs for each advertiser to determine its overall behavior under each consent condition. An advertiser is classified as *Full Consent Compliant* when the *gclid* cookie is set only after consent is accepted (*accept=yes*, *reject=no*, *no-interaction=no*). An advertiser is classified as *Persistent* when the *gclid* cookie is set regardless of the user’s consent choice (*accept=yes*, *reject=yes*), indicating that the advertiser does not condition *gclid* cookie setting on consent. Finally, an advertiser is classified as *Passive* when the *gclid*

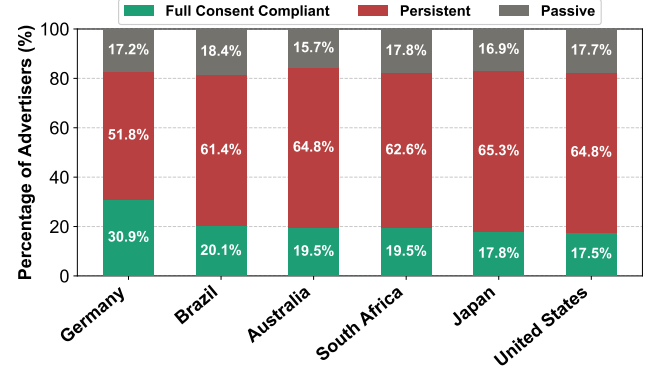


Figure 5: Distribution of advertiser behavior categories across VPs. *Full Consent Compliant* represents advertisers that correctly respect the user’s choice by setting a first-party *gclid* cookie only after consent is accepted and withholding it during both rejection and no-interaction states. *Persistent* behavior identifies advertisers that fail to honor the “Reject All” signal, setting the *gclid* identifier as a first-party cookie even after explicit user rejection. *Passive* behavior includes advertisers that do not set a first-party *gclid* under any circumstances, even when consent is granted.

cookie is not set under any of the three conditions (*accept=no*, *reject=no*, *no-interaction=no*). The reported percentages are computed over advertisers for which consent behavior can be evaluated, i.e., those for which both Accept and Reject interactions are available.

Across all VPs, this group represents a minority of advertisers, ranging from 17.5% (87 out of 497 advertisers) in the United States to 30.9% (185 out of 598 advertisers) in Germany. Notably, even in Germany, where GDPR enforcement is among the strictest globally, fewer than one in three advertisers fully honor the consent signal, suggesting that regulatory pressure alone does not translate into widespread technical compliance. In contrast, the *Persistent* category accounts for the majority of advertisers across all VPs, ranging from 51.8% (310 out of 598 advertisers) in Germany to 65.3% (308 out of 472 advertisers) in Japan. These advertisers set *gclid* as a first-party cookie even after users explicitly reject consent, in direct violation of consent-dependent storage requirements. The consistency of this majority across jurisdictions with markedly different privacy regimes, from Germany to the United States and South Africa, indicates that non-compliance is a structural characteristic of the advertising ecosystem rather than a region-specific anomaly. *Passive* advertisers, who do not set *gclid* under any consent condition, including acceptance, constitute a small and stable fraction, from 15.7% (74 out of 472 advertisers) in Australia to 18.4% (88 out of 477 advertisers) in Brazil. This behavior may reflect an incomplete tracking setup, alternative attribution approaches, or configuration differences.

Beyond per-vantage-point analysis, we examine *globally consistent* behavior, that is, advertisers that exhibit the same category across all VPs. Out of 1,412 advertisers, only 64 advertisers (4.5%) exhibit fully compliant behavior when requiring consistent behavior across all conditions and locations. In contrast, 207 advertisers

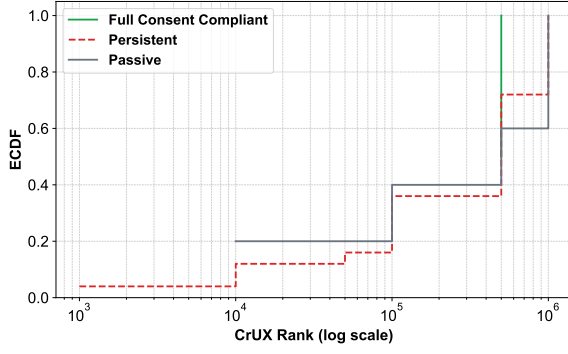


Figure 6: Empirical Cumulative Distribution Functions (ECDF) of CrUX popularity ranks for advertisers exhibiting globally consistent behavior across all vantage points, grouped by behavior category. Lower rank values indicate more popular websites.

(14.7%) persistently set *gclid* regardless of consent state and geographic location. We further examine how these behaviors vary with website popularity using CrUX rankings (Figure 6), where lower rank values indicate more popular websites. The distributions span a broad range of website ranks, with relatively few advertisers associated with the most popular sites. Persistent and Passive advertisers are observed across a wide range of ranks, with their ECDFs increasing substantially at higher rank values. Full Consent Compliant advertisers are concentrated around a relatively high rank value. Overall, the observed behaviors are not predominantly concentrated among the most popular websites but instead extend across a broad range of the web.

We further examine third-party tracking cookie behavior across the three advertiser categories in Figure 7. Specifically, we measure the proportion of advertisers within each category that continue to set third-party cookies after users explicitly reject consent. As expected, advertisers in the *Persistent* category exhibit the highest rates of third-party cookie setting under rejection across all VPs, consistent with their general disregard for consent signals, ranging from 44.5% (138 out of 310) in Germany to 77.6% in the United States and Japan (250 out of 322 and 239 out of 308, respectively). In comparison, *Full Consent Compliant* advertisers generally exhibit lower rates than the *Persistent* category, although the rates remain substantial, ranging from 22.7% (42 out of 185) in Germany to 51.0% (49 out of 96) in Brazil. However, the difference between *Full Consent Compliant* and *Passive* advertisers varies across VPs, with *Full Consent Compliant* advertisers exhibiting slightly higher rates in Brazil and the United States. This cross-mechanism comparison provides additional evidence supporting the behavioral classification and suggests that *Full Consent Compliant* advertisers are less likely than *Persistent* advertisers to set third-party cookies after rejection, although compliance is far from universal. These findings indicate that adherence to consent signals across tracking mechanisms remains incomplete, even among advertisers that otherwise handle *gclid* correctly.

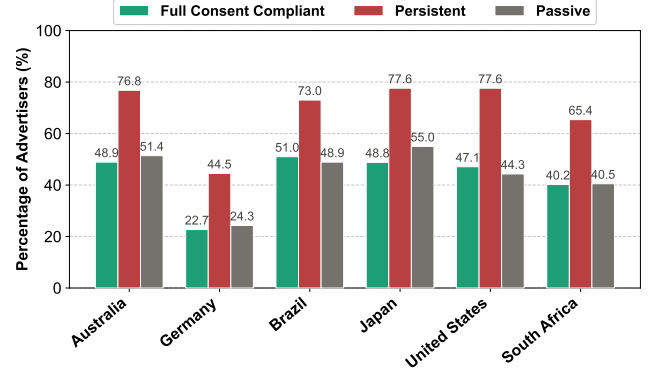


Figure 7: Percentage of advertisers that set third-party cookies after user rejection, grouped by behavior category and vantage point.

Key takeaway: Non-adherence to consent signals is widespread across locations, with most advertisers continuing to track users even after explicit rejection via *gclid*-based tracking. This behavior effectively extends the third-party cookie tracking mechanism.

4.3 *gclid* Parameter Shadowing Analysis

Out of 1,412 advertisers, we detect 19 that employ parameter shadowing to back up *gclid* under alternative parameters (e.g., *ref_id*, *gclidsrc*, *ss_gc*), as summarized in Table 3.

Across browsers, advertiser behavior differs significantly. Chrome and Safari Default browsing show similar behavior, where most of these advertisers successfully set *gclid* as a first-party cookie. In contrast, Safari Private Browsing and Brave enforce stricter protections, preventing *gclid* from being stored for nine and eleven advertisers, respectively. In particular, four advertisers are able to set *gclid* in Chrome and Safari Default mode but fail in Safari Private Browsing and Brave, indicating that these browsers effectively remove the *gclid* parameter before it can be stored. In addition, both browsers block well-known tracking scripts (e.g., Google Tag and Google Analytics), further limiting standard tracking mechanisms.

Interestingly, we identify five advertisers that employ parameter shadowing but do not set *gclid* as a first-party cookie across all tested browsers. This suggests that these advertisers may not rely on client-side cookie storage for tracking. One possible explanation is that the *gclid* value could be captured through URL parameters and processed server-side, for example, via form submissions or backend logging. In such cases, the identifier may be stored and later used for conversion attribution without requiring persistent client-side storage. This behavior could be consistent with server-side or offline conversion tracking mechanisms supported by advertising platforms, where *gclid* values may be uploaded directly for attribution [29]. While we cannot directly observe server-side processing, we believe that parameter shadowing may support tracking strategies that do not depend on browser-based storage.

Furthermore, we find that ten advertisers are able to set *gclid* as a first-party cookie despite the presence of browser-level tracking

Table 3: Behavior on different browsers of advertisers that configure the backup parameter. Note that ✓ represents setting *gclid* as a first-party cookie and ∅ represents *not* setting *gclid* as a first-party cookie.

#	Advertiser	Backup parameter	Chrome	Safari Default	Safari Private	Brave
1	temu.com	_x_ns_gclid	∅	∅	∅	∅
2	wohllivia.de	ref_id	∅	∅	∅	∅
3	peppagame.com	utm_campaign	∅	∅	∅	∅
4	silverpenny.co	aff_unique2	∅	∅	∅	∅
5	thermofisher.com	ef_id	∅	∅	∅	∅
6	redviking.com	delivery_no	✓	✓	∅	∅
7	perspiresaunastudio.com	tid	✓	✓	∅	∅
8	sunvalue.com	s5	✓	✓	∅	∅
9	hyggear.com	utm_term	✓	✓	∅	∅
10	ralphchristian.com	gclid	✓	✓	✓	∅
11	jetsono.com	ref_id	✓	✓	✓	∅
12	makesyoufluent.com	ref_id	✓	✓	✓	✓
13	bannerandflagonline.co.za	gclsrc	✓	✓	✓	✓
14	woodupp.co.za	ss_gc	✓	✓	✓	✓
15	solakon.de	adb_click_id	✓	✓	✓	✓
16	datempire.com	qwerty	✓	✓	✓	✓
17	eyevitaminreviews.com	ref_id	✓	✓	✓	✓
18	hostplus.com.au	ef_id	✓	✓	✓	✓
19	aroundhome.de	click_id	✓	✓	✓	✓

protections. Among them, two advertisers (jetsono.com and ralphchristian.com) set *gclid* as a first-party cookie in Chrome, Safari Default, and Safari Private browsing, but not in Brave. Our analysis shows that, in this case, the Google `gtag.js` script is blocked by Brave’s tracking protections, preventing the site from setting *gclid* via the standard Google tagging mechanism.

In contrast, the remaining eight advertisers successfully set *gclid* as a first-party cookie across all tested browsers, including Brave. We observe that these sites rely on custom JavaScript implementations rather than the standard Google tagging library. Specifically, after the redirection from *AdURLs*, the original *gclid* parameter is removed from the landing URL, consistent with browser-level protections. However, the *gclid* value is preserved in alternative parameters (Table 3), which are not filtered by the browser. These scripts extract the value from the custom parameters and store it as a first-party cookie using client-side code (e.g., `document.cookie`).

Custom identifier implementations are less affected by blocking rules that target known parameter names. As a result, parameter shadowing enables the recovery and persistence of *gclid* even under strict browser protections. Although such cookies are subject to browser-imposed lifetime limitations (e.g., typically up to seven days for JavaScript-set cookies in Safari ITP [51, 52]), this duration may still be sufficient for common attribution windows.

Key takeaway: Advertisers actively circumvent browser-level mechanisms that remove known tracking parameters by using parameter shadowing, aliasing *gclid* within unregulated backup parameters to evade URI redaction and maintain persistent attribution.

4.4 Escalation of *gclid* into Cross-party Leakage

While *gclid* is intended for conversion attribution within a first-party context, its exposure to third-party domains can effectively transform it into a cross-site tracking identifier. Such leakage undermines the isolation provided by first-party storage and enables tracking beyond the original advertiser context. In particular, even when user consent mechanisms restrict cookie storage, the *gclid* identifier may still be transmitted through HTTP requests and shared with external domains.

In this section, we investigate whether and how *gclid* is exposed to third-party tracking domains, and quantify the extent to which this leakage contributes to cross-party tracking.

4.4.1 Impact of user consent to *gclid* leakage. We summarize the average number of third-party tracking domains receiving *gclid* from advertisers in Figure 8. The distributions show broadly consistent patterns across VPs. Under *Accept All*, the median number of third-party tracking domains receiving *gclid* is approximately 5–6 across countries. When users explicitly reject consent, this median generally decreases, ranging from 3.0 to 4.6 domains. Similarly, when users do not interact with the consent banner, the median ranges from 3.8 to 5.0 domains. Thus, rejecting consent reduces third-party *gclid* sharing, but does not eliminate it.

Interestingly, advertisers without a consent banner often exhibit third-party *gclid* sharing levels comparable to those observed under *Accept All*, although the difference is more pronounced in Germany. For example, the median under *No Banner* is close to *Accept All* in Australia (5.8 vs. 6.0), Brazil (5.4 vs. 5.7), Japan (6.0 vs. 6.0), the United States (5.0 vs. 5.25), and South Africa (5.4 vs. 6.0). This pattern suggests that the absence of a consent mechanism is often associated with tracking behavior similar to that observed

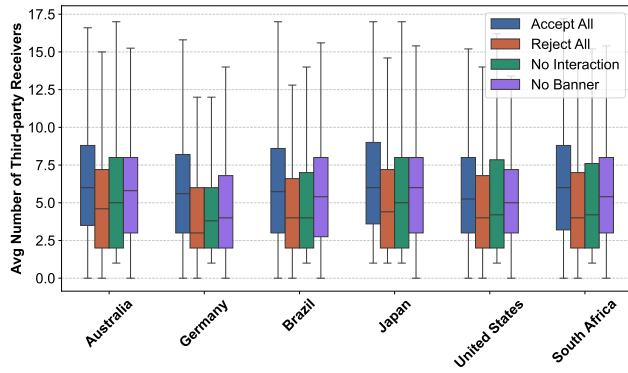


Figure 8: Distribution of unique third-party domains receiving the *gclid* across vantage points. Each data point represents the average number of distinct third-party receivers observed per advertiser over five runs, categorized by consent banner presence and user interaction.

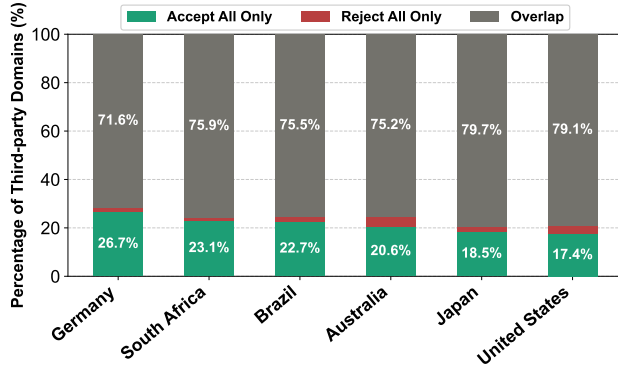


Figure 9: Distribution of third-party tracking domains received *gclid* in Accept All and Reject All interaction.

when users explicitly accept consent, potentially indicating default tracking in the absence of user choice.

We also observe substantial variation within each condition, as reflected by the wide interquartile ranges, standard deviations, and maximum values. Although the median behavior is generally concentrated around 4–6 third-party tracking domains, some advertisers send *gclid* to substantially larger numbers of third-party tracking domains, with observed maxima reaching approximately 31.6–50 domains across conditions and VPs. Overall, the distributions exhibit similar patterns across VPs, with *Accept All* generally associated with the highest third-party sharing and *Reject All* with the lowest, while *No Banner* tends to remain closer to *Accept All* than to *Reject All*.

To further clarify the impact of user interaction on the consent banner, we investigate the distribution of third-party tracking domains when users consent and when users deny consent. Figure 9 shows that advertisers send *gclid* to third-party tracking domains regardless of user consent and country. Between 71.6% of third-party tracking domains (336 out of 469) in Germany and 79.7% of

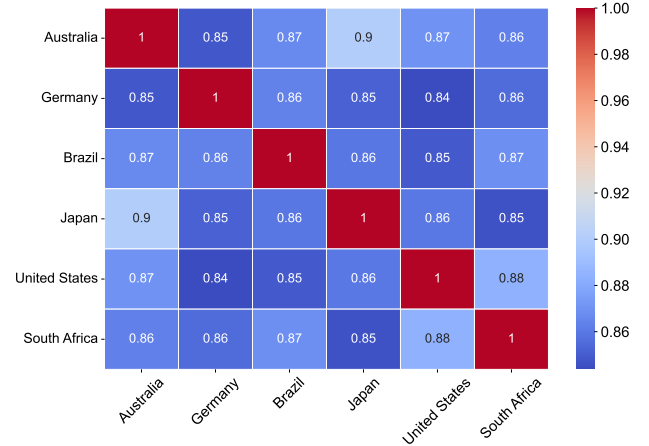


Figure 10: Jaccard similarity between third-party domains received *gclid* across different vantage points (all 5 runs). The heatmap shows the degree of overlap in tracker domains between the VPs, with values closer to 1 indicating higher similarity and values closer to 0 representing lower overlap.

third-party tracking domains (314 out of 394) in Japan receive *gclid* in both cases: when users consent to advertisers collecting data on cookies and when users firmly reject. The percentage of third-party tracking domains that only appear when users accept consent is relatively small, from 17.4% (74 out of 425) in the United States to 26.7% (125 out of 469) in Germany. This suggests that only a limited portion of tracking is actually dependent on user consent. We also observe 58 third-party tracking domains that appear exclusively under rejection across all countries. We list these domains in Table 4 in the Appendix C. Their limited number indicates that rejection rarely introduces distinct tracking behavior; instead, most tracking domains are already active regardless of consent. This behavior likely arises because third-party scripts are loaded before or independently of consent decisions, and consent mechanisms primarily regulate cookie storage rather than network communication. As a result, identifiers such as *gclid* can still be transmitted via URL parameters or headers even when users reject consent.

Key takeaway: While consent banners typically describe the use of cookies, they often fail to explicitly disclose the transmission of user data through third-party network requests. Advertisers send *gclid* to third-party tracking domains regardless of the presence of a consent banner or users’ decision on consent, raising concerns about the actual effectiveness of consent banners.

4.4.2 Cross tracking ecosystem. To further clarify the consistency of tracking behavior across countries, we analyze the similarity between third-party domains received *gclid* across different VPs. We summarize the result in Figure 10. The heatmap shows very high similarity across all countries, with scores ranging from 0.84 to 0.9. This shows that the sets of third-party domains receiving the *gclid* are highly overlapping, regardless of VPs. The highest similarity appears between Australia and Japan (0.9). The lowest similarity,

between Germany and the United States (0.84), is still high, showing the modest difference. Germany tends to have slightly lower similarity scores with other countries, from 0.84 to 0.86, however the gap with the highest score is small.

We breakdown top 10 most popular third-party tracking domains that received *gclid* in Figure 11. Overall, the Google Tag service operates actively, constantly gathering the *gclid* parameter from advertisers. We notice that many other third-party tracking domains, such as Meta (facebook.com), Microsoft Analytics (bing.com, clarity.ms), LinkedIn Conversion Tracking (linkedin.com), and TikTok (tiktok.com), are also attempting to collect *gclid*. This suggests that advertising and tracking platforms are creating a network to share data with each other. This is a sign of cross-platform tracking, where multiple advertising platforms leverage data to understand and track users across the Internet, rather than operating independently. As a result, user data is gathered more extensively than users realize, even when they only interact with a single advertising platform.

Key takeaway: There is an exchange of *gclid* with the network of trackers and advertisers ecosystem, creating a process of user profile synchronization.

5 DISCUSSION

In this section, we discuss factors that may contribute to the observed non-adherence to consent signals, focusing on implementation challenges in Google Consent Mode, and outline limitations and directions for future work.

5.1 Implementation Challenges in Google Consent Mode

We believe that the prevalence of *Persistent* behavior is not necessarily the result of deliberate circumvention by advertisers, but may reflect a fundamental architectural property of Google's Consent Mode framework. Specifically, Google's official documentation states that, by default, no consent mode values are set [28]. When an advertiser deploys the Google tag without explicitly invoking the consent default API, Google tags fire unconditionally, regardless of the user's choice in the cookie banner. The consent banner and the Google tag infrastructure operate as two independent systems: the former captures the user's choice, while the latter remains unaware of it unless the advertiser explicitly connects the two.

Even when advertisers invoke the consent API, correct configuration is non-trivial. Each consent parameter must be explicitly set to denied before any measurement data is sent, as there is no privacy safe default. Google's recommended implementation requires the following declaration to precede any tag execution:

```
gtag('consent', 'default', {
  'ad_storage': 'denied',
  'ad_user_data': 'denied',
  'ad_personalization': 'denied',
  'analytics_storage': 'denied'
});
```

Furthermore, if the consent default call is made after the Google tag has begun executing, the defaults will not take effect [28]. This

creates an additional failure mode in which advertisers who intend to implement consent mode correctly may still fail due to timing issues between tag loading and consent initialization, especially when consent management platforms load asynchronously. Beyond the first-party *gclid* cookie, the framework's limitations extend to ad click data more broadly. Even when *ad_storage* is set to denied, Google's documentation states that data sent to Google may still include the full page URL, including ad click information such as the *gclid* parameter. Advertisers must additionally enable a separate *ads_data_redaction* setting to suppress this data, introducing another configuration step.

Taken together, these architectural choices, including the absence of a privacy safe default, strict ordering requirements, and multiple configuration steps, shift compliance responsibility to advertisers. As a result, full consent compliance requires deliberate and correct implementation, while non-compliance may occur by default. Similar to prior work, which finds that Google encourages the use of more invasive data collection by recommending the least private default configurations during setup [34], our results suggest that system design may contribute to widespread non-adherence to consent signals. This observation has important implications for regulatory interpretation and enforcement. Under the GDPR and the ePrivacy Directive, the legal obligation to obtain prior consent before setting non-essential cookies falls on the data controller, namely the advertiser, rather than the platform providing the tracking infrastructure [8, 22]. While Google's design does not remove this responsibility, it highlights a structural tension in which the dominant tracking infrastructure does not enforce consent by default. When non-compliance is the path of least resistance, the gap between legal requirements and practical implementation is likely to persist.

5.2 Advertiser Jurisdiction and Cross-Border Compliance

Our measurements primarily vary the geographic location of users, since the applicability of privacy requirements can depend on the jurisdiction of the individuals whose data are processed. For example, Article 3(2) of the GDPR can extend its applicability to organizations outside the EU when their processing activities involve offering goods or services to individuals in the EU or monitoring their behavior. Thus, the physical location of an advertiser alone does not determine whether GDPR obligations apply. In our study, using multiple vantage points allows us to examine how the same advertiser behaves when accessed by users from different jurisdictions.

At the same time, incorporating advertiser jurisdiction would provide an additional perspective on cross-border compliance. However, determining an advertiser's relevant jurisdiction is non-trivial. An advertiser may operate through multiple legal entities, serve users across different countries, or use third-party advertising and analytics services whose infrastructure and data-processing entities are located in different jurisdictions. Moreover, the domain hosting an advertisement or the location of the advertiser's web server may not correspond to the legal entity responsible for processing the user's data. Consequently, identifying the jurisdiction that is legally relevant to a particular tracking practice would require information beyond what can be directly observed from our web measurements,

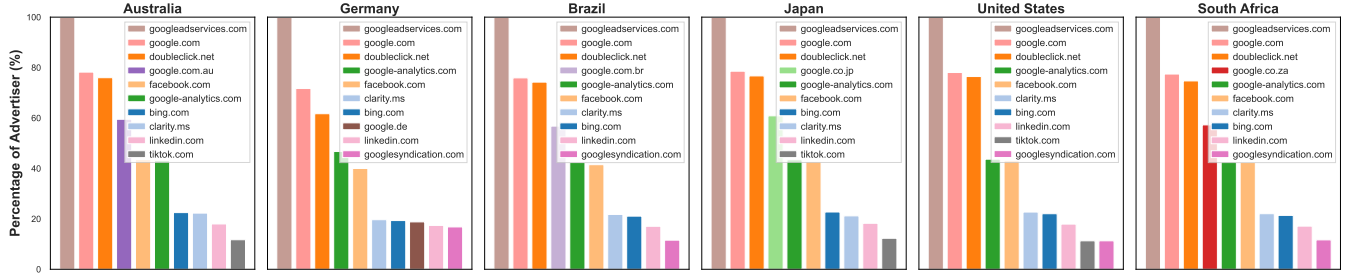


Figure 11: Top 10 most popular third-party tracking domains received *gclid* and their prevalence across different VPs.

such as the advertiser’s corporate structure, applicable terms and policies, and relationships with third-party data processors.

Cross-border enforcement presents a related challenge. Even when a tracking practice is subject to a user’s local privacy requirements, investigating or enforcing potential violations may involve an organization established in another jurisdiction, potentially with additional entities or service providers involved in the data-processing chain. Therefore, geographic differences in our measurements should not be interpreted as direct evidence of differences in advertisers’ legal compliance. We believe that systematically linking advertisers to their relevant legal jurisdictions, identifying the entities responsible for data processing, and examining how these factors relate to observed consent behavior would be a valuable direction for future work.

5.3 Limitations and Future Work

Our study has several limitations. First, our measurements rely on automated browsing using controlled environments and VPN-based VPs. While this allows for reproducibility and cross-location comparison, it may not fully capture real user behavior, such as long-term browsing patterns, logged-in states, or personalized advertising experiences. In addition, our use of commercial VPN as geographic vantage points introduces a potential measurement bias. Recent work has shown that VPN-based Web measurements can vary substantially across VPN providers even within the same country, due to differences in DNS resolution, CDN routing, and network paths that affect which endpoints and replicas are reached [48]. In our setting, VPN or datacenter IP addresses may also receive different treatment from those of residential users. Therefore, although our controlled VPN-based setup enables reproducible cross-country comparisons, we cannot completely isolate jurisdictional effects from potential VPN-induced differences. We consequently interpret the observed geographic differences, including the lower persistence rate observed in Germany, as suggestive rather than causal evidence of jurisdictional effects. Future work should complement VPN-based measurements with residential or ISP-based vantage points to validate these geographic patterns.

Second, we focus on the *gclid* identifier as a representative case of URL-based tracking; other identifiers (e.g., *fbclid*, *msclkid*) may exhibit different behaviors. Finally, our study evaluates adherence to consent signals based on observed behavior but cannot determine the intent or internal implementation choices of advertisers. For

future work, we plan to extend our analysis to other identifiers, combine measurement with controlled experiments or collaborations with platform providers to better understand the root causes of observed discrepancies, and investigate server-side tracking mechanisms.

6 Conclusion

In this paper, we present a comprehensive measurement of *gclid*-based tracking on the web under controlled consent conditions and across multiple geographic locations. Our findings show that *gclid*-based tracking is pervasive and often persists regardless of user consent, with many advertisers continuing to store or transmit identifiers even after explicit rejection. We further observe that this behavior extends across tracking mechanisms and that advertisers can adapt to browser-level protections through techniques such as parameter shadowing. In addition, *gclid* is widely shared with third-party domains, forming a consistent tracking ecosystem across regions. Overall, these results highlight a gap between the intended behavior of consent mechanisms and the observed persistence of identifier-based tracking in practice. We hope this work contributes to a better understanding of tracking systems on the web and informs the development of more effective privacy protections.

Ethical Considerations

Our study does not involve human subjects, surveys, or user studies. All data was collected from publicly accessible advertiser landing pages through automated crawling, simulating the behaviour of a regular web user visiting a Google Ads landing page. Our study investigates *gclid*-based tracking that is invisible to passive observation, and there is no alternative methodology to measure them without collecting *AdURLs* and visiting the resulting landing pages.

We acknowledge that visiting *AdURLs* through automated crawling may register as clicks on Google Ads, potentially resulting in pay-per-click charges to advertisers. However, precisely quantifying these costs is challenging, as advertisements can be charged in different ways, such as cost-per-click (CPC), cost-per-thousand views (CPM), or cost-per-action (CPA), and costs vary depending on the publisher and placement. We do not complete any conversion actions such as purchases or form submissions, and the number of visits per advertiser across our measurement rounds is small. We therefore consider the financial impact of our crawls on advertisers to be negligible, consistent with prior measurement studies that

similarly engaged with ad landing pages to analyze online advertising and tracking practices [14, 17, 42]. The digital advertising industry is a trillion-dollar ecosystem driven largely by user profiling and targeting. Academic research that directly engages with ad systems to measure practices such as *gclid*-based tracking, which operates across consent boundaries and browser-level privacy protections, is crucial for consumers, regulators, and developers of privacy-enhancing technologies. We believe these benefits far outweigh any minor costs incurred. We will make our analysis scripts, raw dataset, and findings publicly available upon publication to support reproducibility and further research in the community.

Open Science

This study adheres to the principles of transparency and reproducibility promoted by the open science policy. To support these principles, we shared an open replication package containing our crawler, raw data, analysis scripts, and visualizations: <https://github.com/fukuda-lab/GCLIDTracking>

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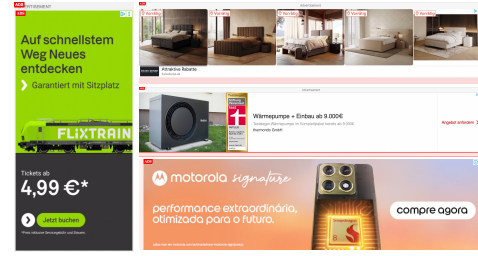
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A BannerClick Performance

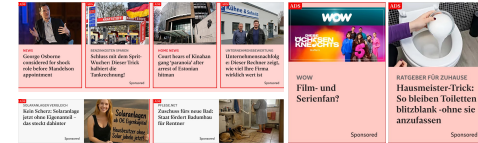
To further assess the performance of *BannerClick* for banner interaction, we run this function automatically on 50 news websites, to auto-detect and interact on cookie consent if detected. We also took screenshots of the consent banner, interaction buttons, and text on these buttons. Then we manually visit these websites to verify the performance of *BannerClick*. Out of 50 sites, *BannerClick* was successfully clickable on 47 sites for “Accept all” (94%); 45 sites were clickable for “Reject all” (90%) due to the reject option being hidden on five sites. These performances further highlight the strength and consistency of *BannerClick*.

B AdURLs Detection Performance

We evaluate the effectiveness of *AdsDetector* by applying it to 20 popular news websites and manually inspecting screenshots of both the homepage and inner pages. Across these websites, we confirm that *AdsDetector* successfully identifies both display ads (Figure 12a) and native ads (Figure 12b). However, we observe a small number of missed cases. Specifically, one website employs bot detection that prevents page loading, and two websites embed native advertisements served by lesser-known or unsupported platforms, which are not detected by our system. Given that our study focuses on *gclid*-based tracking in Google Ads, these limitations do not affect our analysis, as most relevant advertisements are successfully detected.



(a) Detected display ads.



(b) Detected native ads.

Figure 12: Examples of detected advertisements. (a) Display ads and (b) native ads identified by *AdsDetector*. The red “ADS” labels are added by our crawler to indicate detected ads.

C Third-Party Domains Receiving *gclid* Only Under Rejection

Table 4 presents a list of third-party domains that receive the *gclid* identifier exclusively under the *Reject* condition, i.e., domains observed after user rejection but not under acceptance.

Table 4: Third-party tracking domains that receive *gclid* exclusively under the *Reject* condition across all VPs.

Country	Third-party domains	Country	Third-party domains
AU	akamaihd.net	BR	lomadee.com.br
AU	appsflyer.com	BR	marketo.com
AU	attentivemobile.com	BR	xiaoman.cn
AU	bazaarvoice.com	JP	affirm.com
AU	datadoghq-browser-agent.com	JP	akamaihd.net
AU	dv.tech	JP	bugsnag.com
AU	getsmartcontent.com	JP	grafana.net
AU	im-apps.net	JP	onetrust.com
AU	invoca.net	JP	salecycle.com
AU	liadm.com	JP	smartnews-ads.com
AU	metricool.com	US	basis.net
AU	onetag-sys.com	US	clicktripz.com
AU	requestmetrics.com	US	dstillery.com
AU	rtbhouse.com	US	getsmartcontent.com
AU	secureserver.net	US	grafana.net
AU	youtube.com	US	imrworldwide.com
AU	zineone.com	US	koddi.com
DE	affirm.com	US	lijit.com
DE	betterstackdata.com	US	marker.io
DE	metricool.com	US	media6degrees.com
DE	newsbreak.com	US	mindtrip.ai
DE	noibu.com	US	quantcount.com
DE	onetrust.com	US	quantserve.com
DE	srvmath.com	US	youtube.com
DE	trendemon.com	US	zprk.io
BR	abgimea.com	ZA	grafana.net
BR	azure.com	ZA	hlx.page
BR	im-apps.net	ZA	metricool.com
BR	imrworldwide.com	ZA	smartnews-ads.com